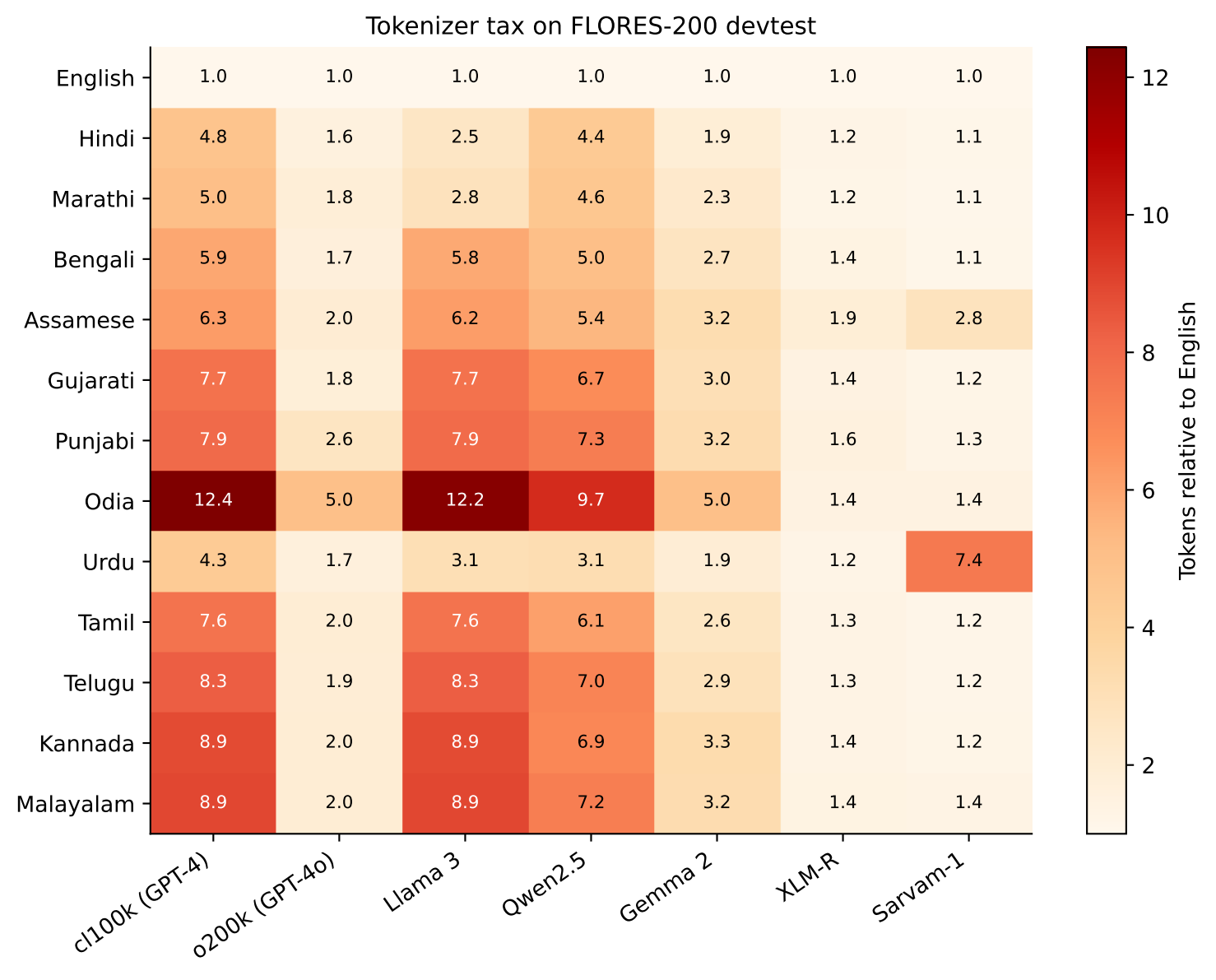
Tokenizer tax on FLORES-200 devtest
English 1.0 1.0 1.0 1.0 1.0 1.0 1.0
Hindi 4.8 1.6 2.5 4.4 1.9 1.2 1.1
Marathi 5.0 1.8 2.8 4.6 2.3 1.2 1.1
Bengali 5.9 1.7 5.8 5.0 2.7 1.4 1.1
Assamese 6.3 2.0 6.2 5.4 3.2 1.9 2.8
Gujarati 7.7 1.8 7.7 6.7 3.0 1.4 1.2
Punjabi 7.9 2.6 7.9 7.3 3.2 1.6 1.3
Odia 12.4 5.0 12.2 9.7 5.0 1.4 1.4
Urdu 4.3 1.7 3.1 3.1 1.9 1.2 7.4
Tamil 7.6 2.0 7.6 6.1 2.6 1.3 1.2
Telugu 8.3 1.9 8.3 7.0 2.9 1.3 1.2
Kannada 8.9 2.0 8.9 6.9 3.3 1.4 1.2
Malayalam 8.9 2.0 8.9 7.2 3.2 1.4 1.4
cl100k (GPT-4)
o200k (GPT-4o)
Llama 3
Qwen2.5
Gemma 2
XLM-R
Sarvam-1
Tokens relative to English
2
4
6
8
10
12